

Cloud Watch

Steps to generate Alarm

Launch the Instance

Go to EC2

Launch Instance

Select the below option which is marked in red.

1. Choose AMI2. Choose Instance Type3. Configure Instance4. Add Storage5. Add Tags6. Configure Security Group7. Review

Cancel and Exit

Step 1: Choose an Amazon Machine Image (AMI)

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. You can select an AMI provided by AWS, our user community, or the AWS Marketplace; or you can select one of your own AMIs.

Q Search for an AMI by entering a search term e.g. "Windows"

Quick Start

My AMIs

AWS Marketplace

Community AMIs

Amazon Linux

Free tier eligible

Amazon Linux 2 AMI (HVM), SSD Volume Type - ami-0c9d48b5db609ad6e

Amazon Linux 2 comes with five years support. It provides Linux kernel 4.14 tuned for optimal performance on Amazon EC2, systemd 219, GCC 7.3, Glibc 2.26, Binutils 2.29.1, and the latest software packages through extras.

64-bit (x86)

Select

Root device type: ebsVirtualization type: nvmm

1 to 36 of 36 AMIs

computing needs.

Filter by: All instance typesCurrent generationShow/Hide Columns

Currently selected: t2.micro (Variable ECUs, 1 vCPUs, 2.5 GHz, Intel Xeon Family, 1 GiB memory, EBS only)

	Family	Type	vCPUs	Memory (GiB)	Instance Storage (GB)	EBS-Optimized Available	Network Performance	IPv6 Support
☐	General purpose	t2.nano	1	0.5	EBS only	-	Low to Moderate	Yes
☒	General purpose	t2.micro Free tier eligible	1	1	EBS only	-	Low to Moderate	Yes
☐	General purpose	t2.small	1	2	EBS only	-	Low to Moderate	Yes
☐	General purpose	t2.medium	2	4	EBS only	-	Low to Moderate	Yes

Cancel

Previous

Review and Launch

Next: Configure Instance Details

Click on Configure instance details

aws Services Resource Groups

1. Choose AMI 2. Choose Instance Type 3. Configure Instance 4. Add Storage 5. Add Tags 6. Configure Security Group 7. Review

Step 3: Configure Instance Details

Configure the instance to suit your requirements. You can launch multiple instances from the same AMI, request Spot instances to take advantage of the lower pricing, assign an access role to the instance, and more.

Number of instances Launch into Auto Scaling Group

Purchasing option ☐ Request Spot instances

Network Create new VPC

Subnet Create new subnet
11 IP Addresses available

Auto-assign Public IP

Placement group ☐ Add instance to placement group.

Enter the details of Network (VPC)

Subnet (auto selection) and enable the Public IP as in the above screen shot.

Then click on add storage

Step 4: Add Storage

Your instance will be launched with the following storage device settings. You can attach additional EBS volumes and instance store volumes to your instance, or edit the settings of the root volume. You can also attach additional EBS volumes after launching an instance, but not instance store volumes. [Learn more](#) about storage options in Amazon EC2.

Volume Type	Device	Snapshot	Size (GiB)	Volume Type	IOPS	Throughput (MB/s)	Delete on Termination	Encrypted
Root	/dev/xvda	snap-0a970fd117e0ad93a	8	General Purpose SSD (gp2)	100 / 3000	N/A	☒	Not Encrypted

Add New Volume

Free tier eligible customers can get up to 30 GB of EBS General Purpose (SSD) or Magnetic storage. [Learn more](#) about free usage tier eligibility and usage restrictions.

Cancel Previous **Review and Launch** Next: Add Tags

Click on Add tag (nothing to be enter in the above screen)

Step 5: Add Tags

A tag consists of a case-sensitive key-value pair. For example, you could define a tag with key = Name and value = Webserver. A copy of a tag can be applied to volumes, instances or both. Tags will be applied to all instances and volumes. [Learn more](#) about tagging your Amazon EC2 resources.

Key (127 characters maximum)	Value (255 characters maximum)	Instances	Volumes
------------------------------	--------------------------------	-----------	---------

This resource currently has no tags

Choose the Add tag button or [click to add a Name tag](#).
Make sure your [IAM policy](#) includes permissions to create tags.

Add Tag (Up to 50 tags maximum)

Cancel Previous **Review and Launch** Next: Configure Security Group

Click on Configure Security Group

A security group is a set of firewall rules that control the traffic for your instance. On this page, you can add rules to allow specific traffic to reach your instance. For example, if you want to set up a web server and allow Internet traffic to reach your instance, add rules that allow unrestricted access to the HTTP and HTTPS ports. You can create a new security group or select from an existing one below. [Learn more](#) about Amazon EC2 security groups.

Assign a security group: ☒ Create a new security group
☐ Select an existing security group

Security group name:
Description:

Type ⁱ	Protocol ⁱ	Port Range ⁱ	Source ⁱ	Description ⁱ
SSH	TCP	22	Custom 0.0.0.0/0	e.g. SSH for Admin Desktop

Add Rule



Warning

Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

[Cancel](#) [Previous](#) [Review and Launch](#)

Click on Review and launch

**Amazon Linux 2 AMI (HVM), SSD Volume Type - ami-0c9d48b5db609ad6e**
Free tier eligible Amazon Linux 2 comes with five years support. It provides Linux kernel 4.14 tuned for optimal performance on Amazon EC2, systemd 219, GCC 7.3, Glibc 2.26, Binutils 2.29.1, and the latest software packages through extras.
Root Device Type: ebs Virtualization type: hvm

▼ Instance Type

[Edit instance type](#)

Instance Type	ECUs	vCPUs	Memory (GiB)	Instance Storage (GB)	EBS-Optimized Available	Network Performance
t2.micro	Variable	1	1	EBS only	-	Low to Moderate

▼ Security Groups

[Edit security groups](#)

Security group name	launch-wizard-1
Description	launch-wizard-1 created 2019-01-19T21:57:09.783+05:30

[Cancel](#) [Previous](#) [Launch](#)

Click on Lanuch

Select an existing key pair or create a new key pair

A key pair consists of a **public key** that AWS stores, and a **private key file** that you store. Together, they allow you to connect to your instance securely. For Windows AMIs, the private key file is required to obtain the password used to log into your instance. For Linux AMIs, the private key file allows you to securely SSH into your instance.

Note: The selected key pair will be added to the set of keys authorized for this instance. Learn more about [removing existing key pairs from a public AMI](#).

Choose an existing key pair

Select a key pair

abc1

☒ I acknowledge that I have access to the selected private key file (abc1.pem), and that without this file, I won't be able to log into my instance.

Cancel

Launch Instances

Choose the existing pair key or create the new one

Then click on Launch Instances

aws

Services

Resource Groups

Kavithab

Sydney

Launch Status

✓

Your instances are now launching

The following instance launches have been initiated: **i-0223afabcfbc1a74a** [View launch log](#)

i

Get notified of estimated charges

Create billing alerts to get an email notification when estimated charges on your AWS bill exceed an amount you define (for example, if you exceed the free usage tier).

Click on the id which is marked in the above screen

Launch Instance

Connect

Actions

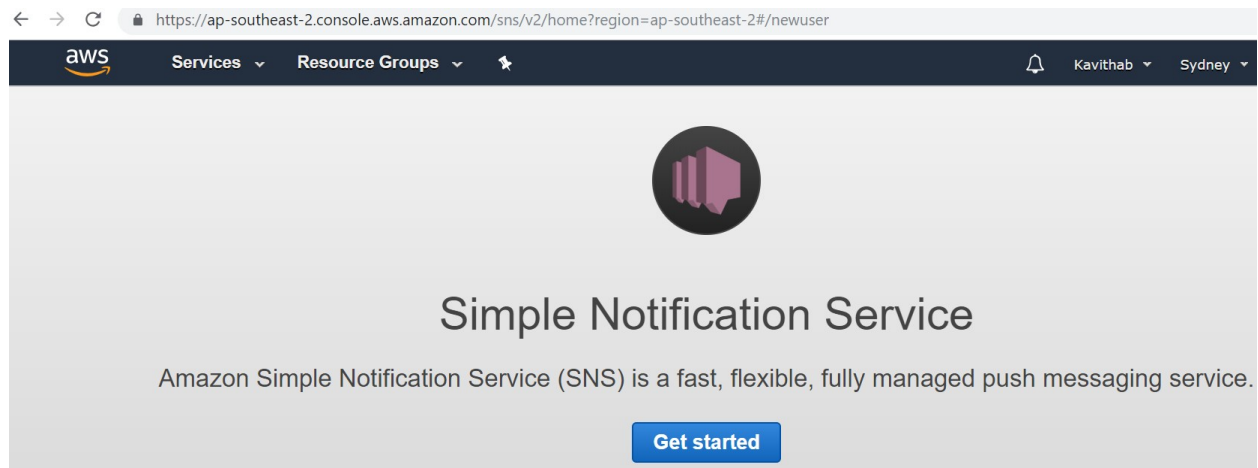
search : i-0223afabcfbc1a74a

Add filter

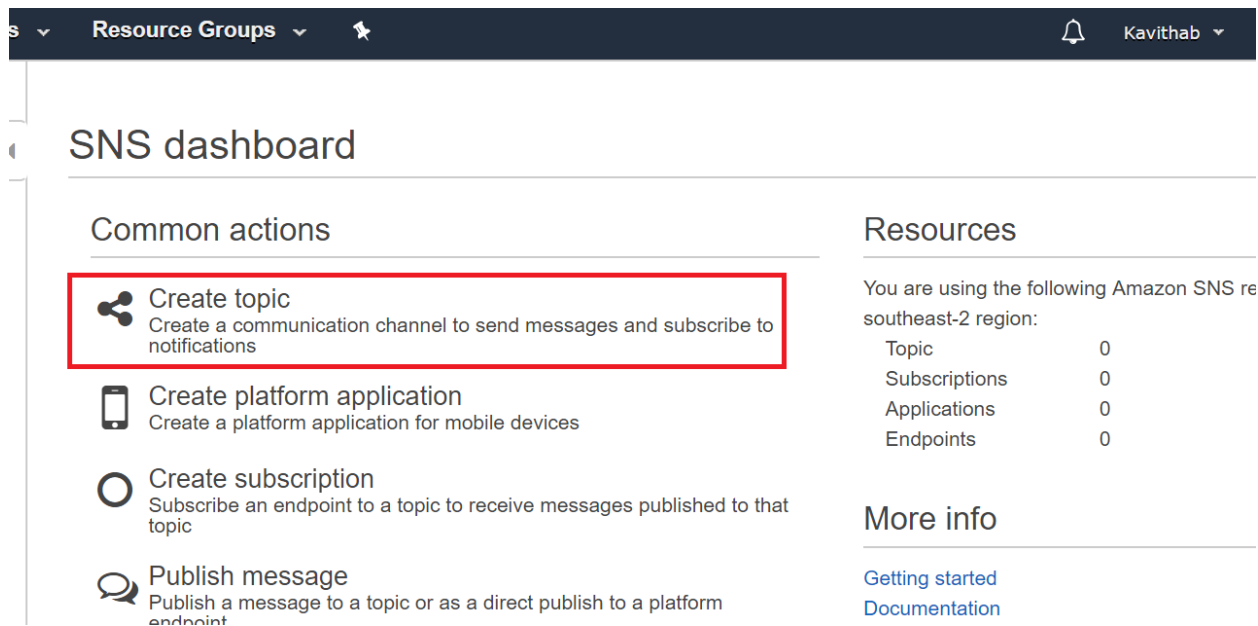
	Name	Instance ID	Instance Type	Availability Zone	Instance State	Status Checks	Alarm Status	Public DNS
	ABC1	i-0223afabcfbc1a74a	t2.micro	ap-southeast-2a	running	2/2 checks ...	None	ec2-52-65-111

Your instance is Launched now.

From the services select Simple Notification Services



Click on Get Started



Click on Create Topic

Create new topic

A topic name will be used to create a permanent unique identifier called an Amazon Resource Name (ARN).

Topic name ⓘ

Display name ⓘ

[Cancel](#) [Create topic](#)

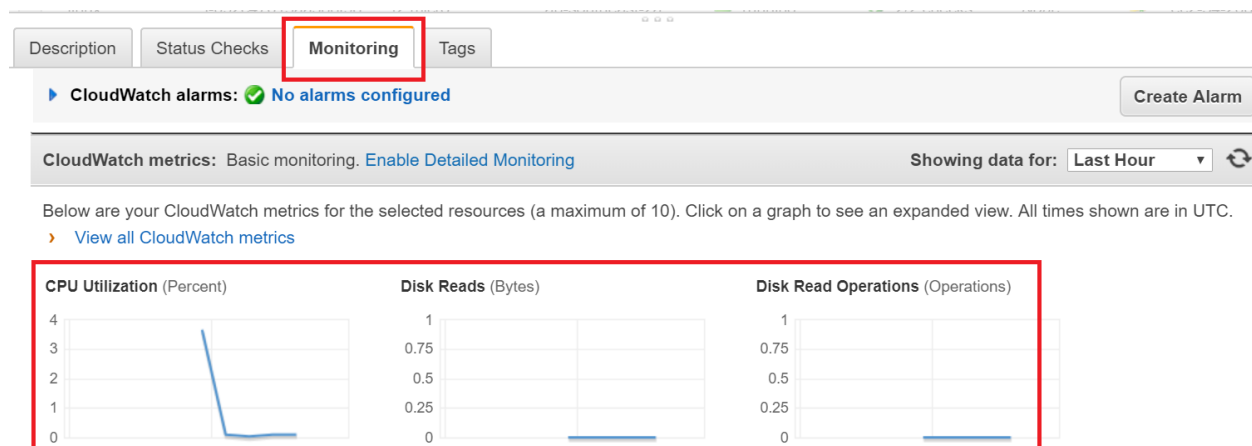
Enter the Topic name

Display name (Incident topic) In the above screen shot we are generating alarm for CPU Utilization, set display name as CPU Spike

Then click on Create Topic

Go to Instance

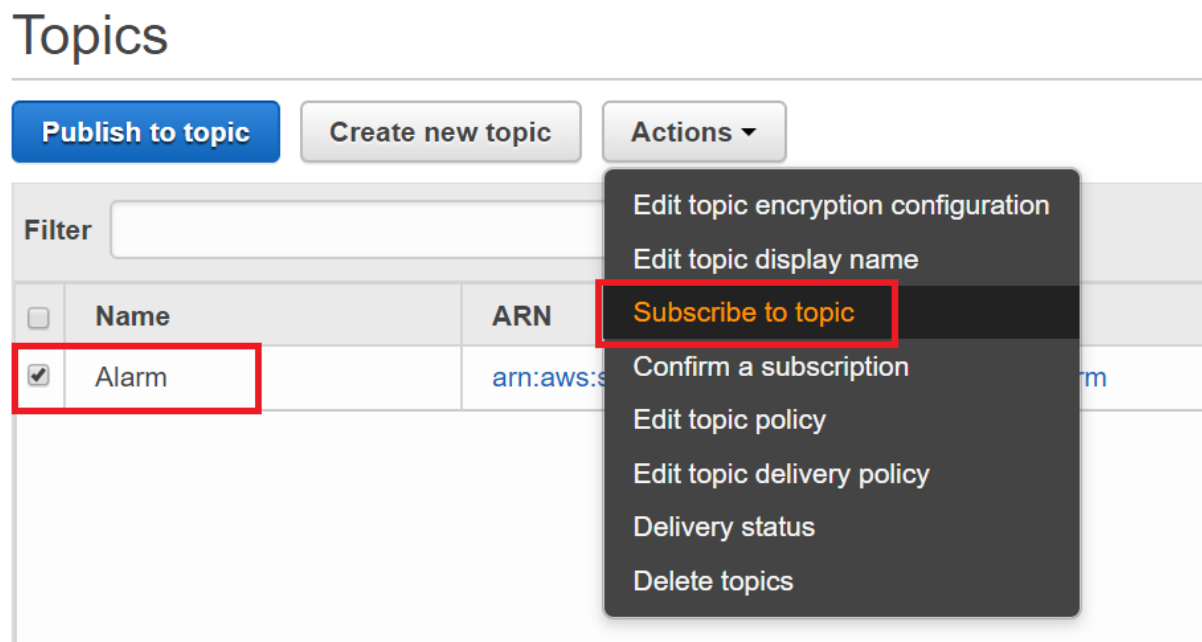
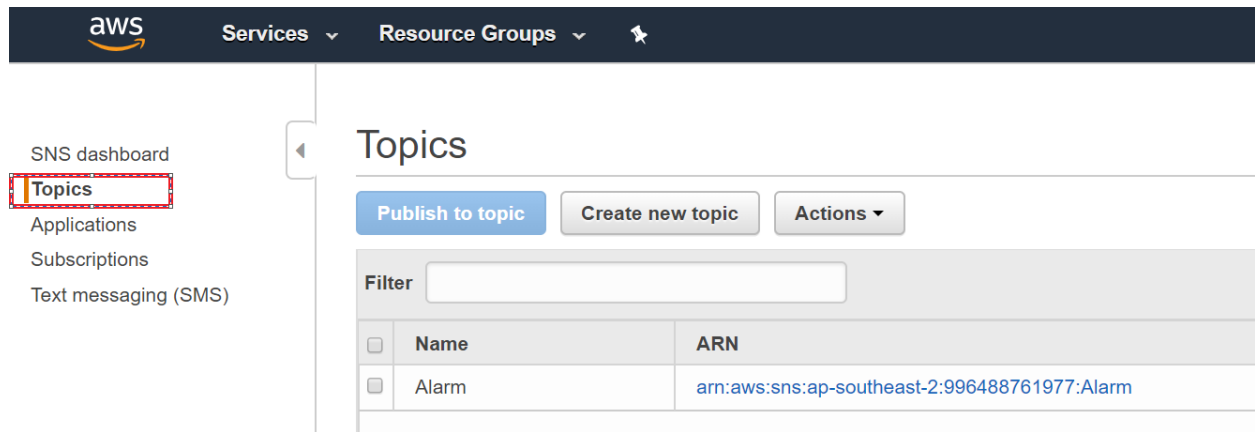
Click on Monitoring



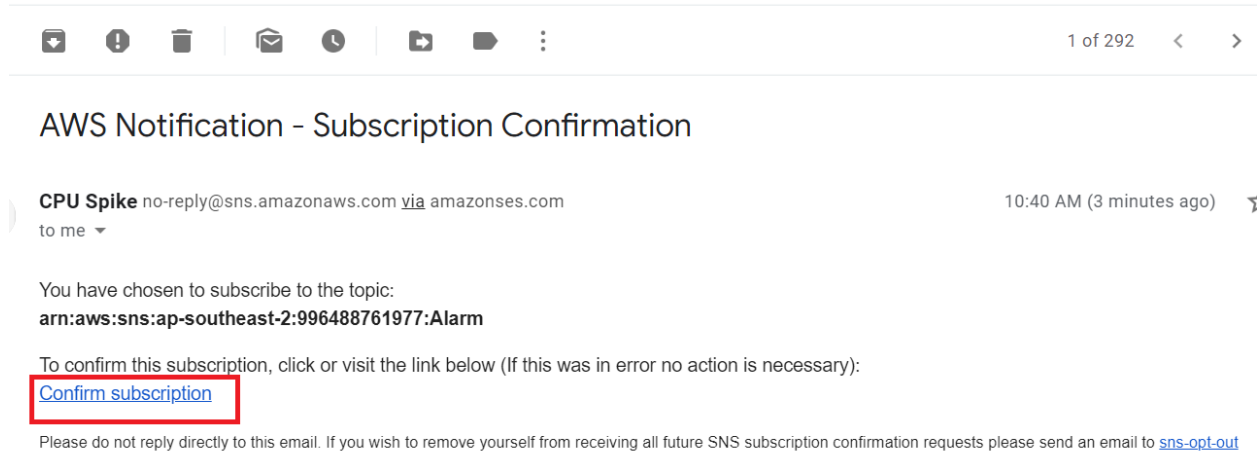
We can view the Cpu utilization details

Then go to SNS (simple notification services)

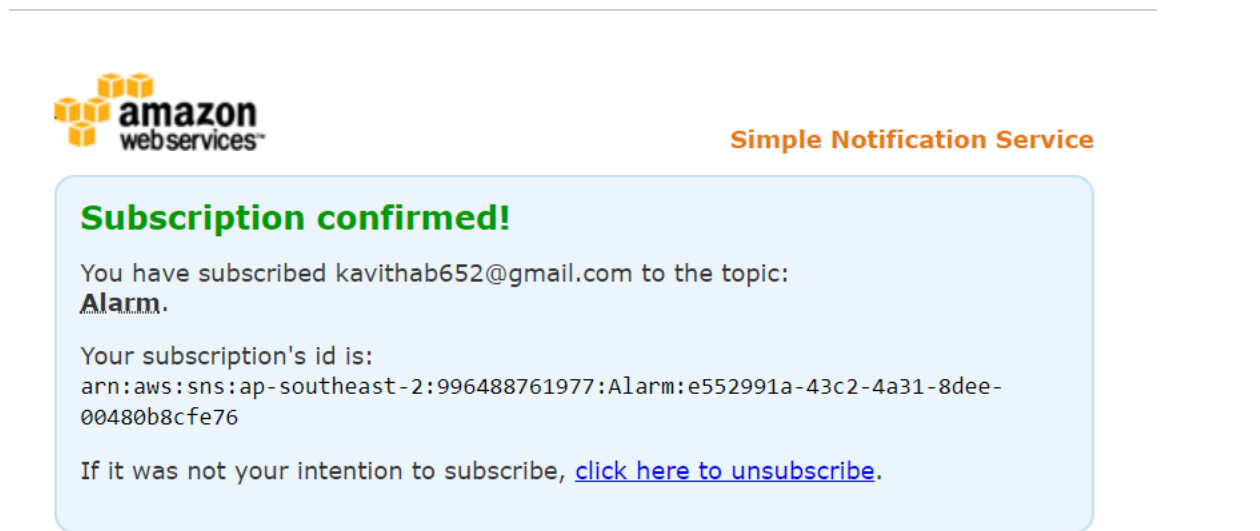
Click on Topics



Select your topic and from Actions select Subscribe topic



Open the notification mail and click on Confirm Subscription



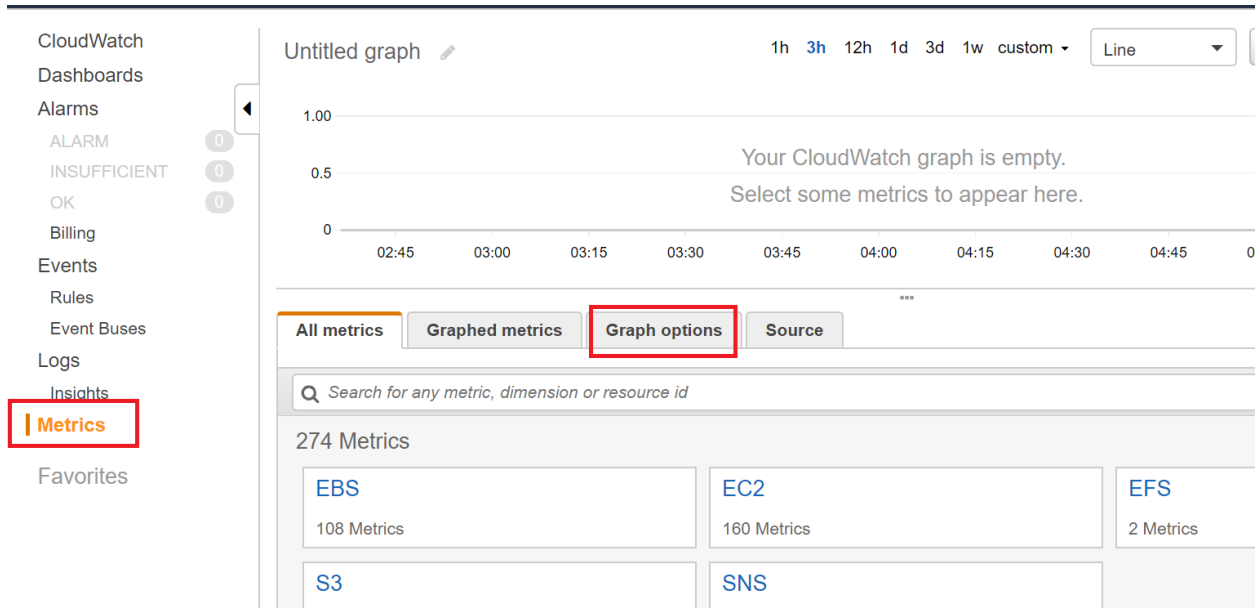
Then go to SNS and click on Refresh button



Then go to services

Select Cloud Watch

Then click on Metrics



Copy the instance ID from Instance

Launch Instance ▼

Connect

Actions ▼

Filter by tags and attributes or search by keyword

☐	Name ▼	Instance ID ▲	Instance Type ▼	Avai
☒	Alarm	i-066a2a0deb46760...	t2.micro	ap-sc
☐	linux	i-092c47613aa388f98	t2.micro	ap-sc

Instance: **i-066a2a0deb4676099 (Alarm)** Public DNS: ec2-54-252

Description

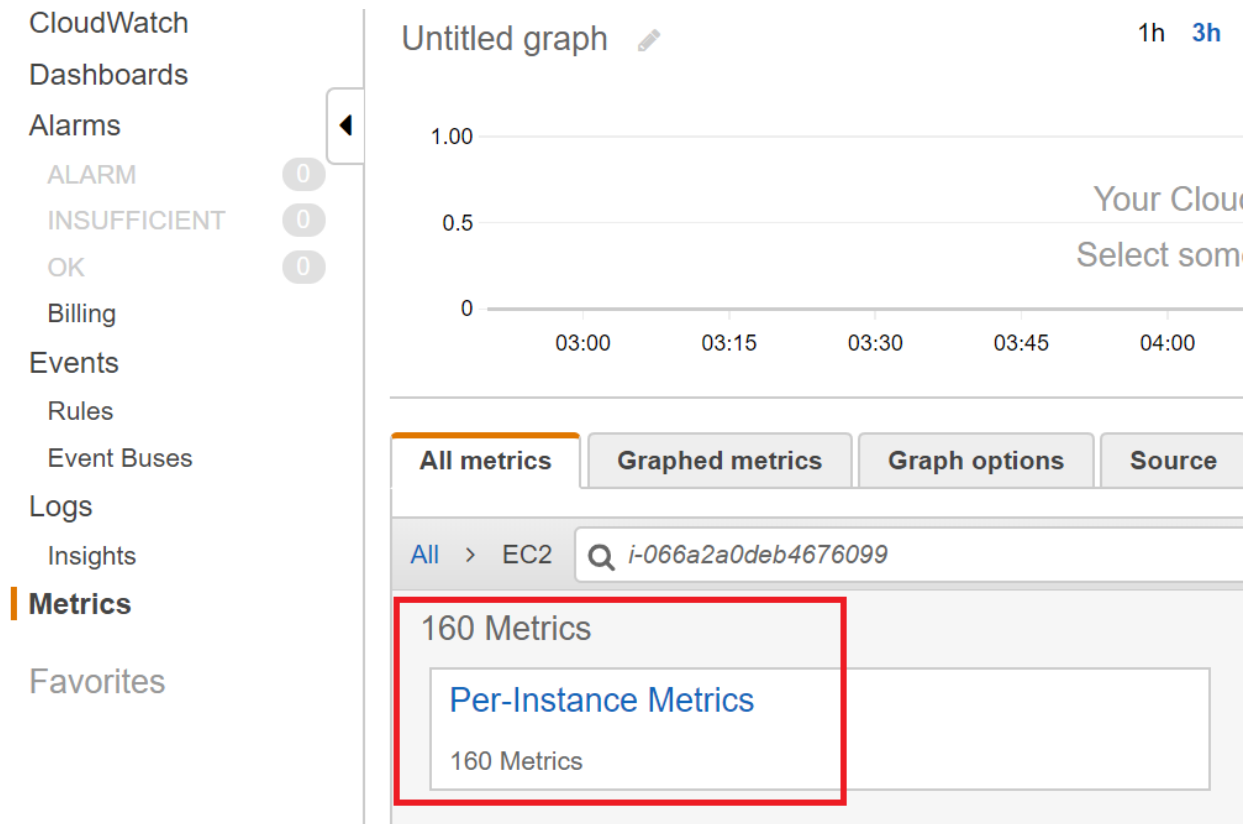
Status Checks

Monitoring

Tags

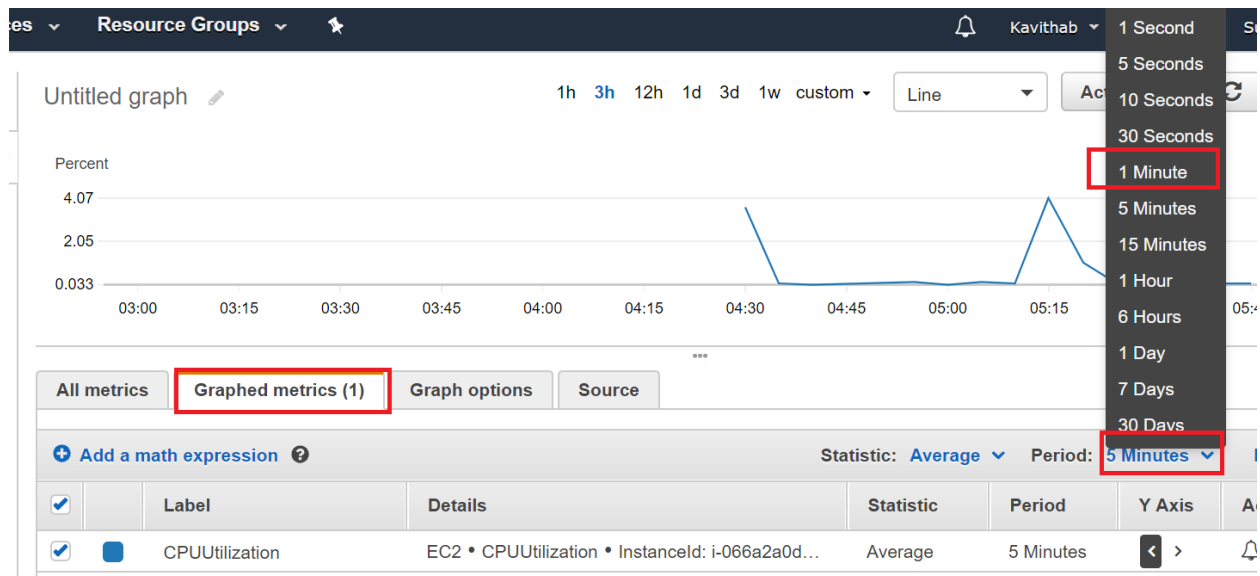
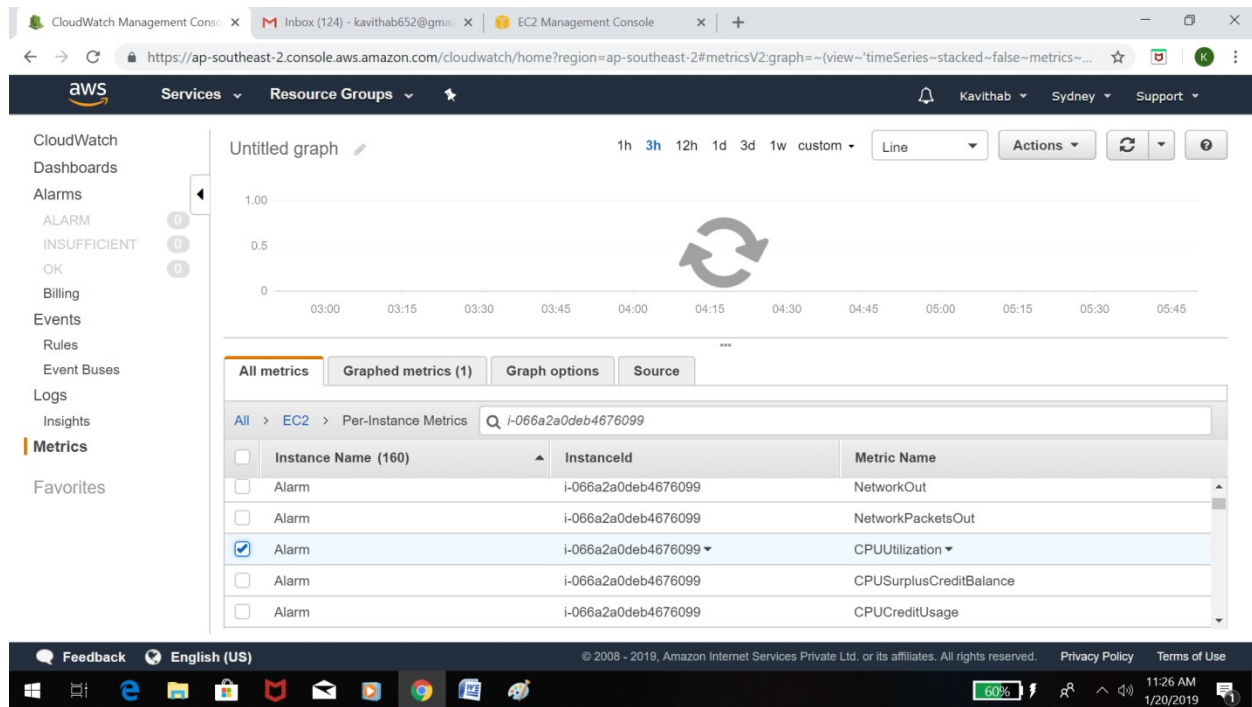
Instance ID **i-066a2a0deb4676099**

Then go to Clouc watch ->Metrics -> EC2---> Pre instance Metrics



Then copy and paste the Instance id

Then select the Alarm of CPU Utilization from the given list below



Select Graphed Metrics -> Drop down the Time Period - we can set the time as per our requirement

In the above screen set for 1 minute

Alarm details

Provide the details and threshold for your alarm. Use the graph to help set the appropriate threshold.

Name:

Description:

Whenever: CPUUtilization

is:

for: 1 out of 1 datapoints

Additional settings

Provide additional configuration for your alarm.

Treat missing data as:

Select the Send notification to as Alarm from the drop down list

Additional settings

Provide additional configuration for your alarm.

Treat missing data as:

Actions

Define what actions are taken when your alarm changes state.

Notification Delete

Whenever this alarm:

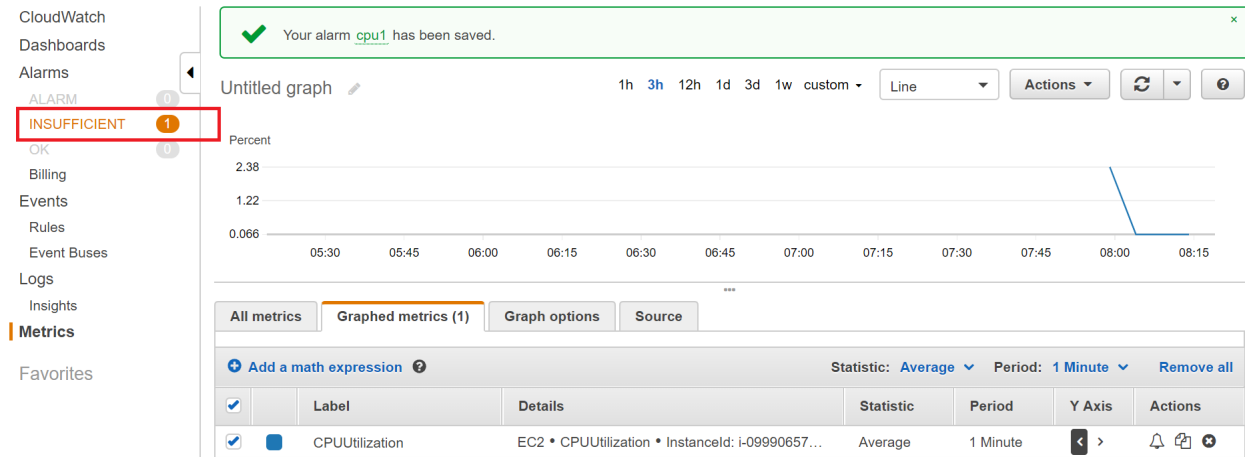
Send notification to: New list Enter list

Email list:

+ Notification + AutoScaling Action + EC2 Action

Cancel **Create Alarm**

Then Click on Create Alarm



Now we have to enable the detailed Monitoring

Then go to Instance Right click on the Enable Detailed Monitoring to enable the

Launch Instance Connect Actions

Filter by tags and attributes or search by keyword

☒	Name	Instance ID	Instance Type	Availability Zone	Instance State	Status Checks
☒	linux	i-099906574151041...	t2.micro	ap-southeast-2a	running	2/2 checks ...

Instance: **i-09990657415104108 (linux)** Public DNS: `ec2-13-210-109-155.ap-southeast-2.compute.amazonaws.com`

Description Status Checks **Monitoring** Tags

▶ CloudWatch alarms: 1 of 1 in **INSUFFICIENT DATA**

CloudWatch metrics: Basic monitoring **Enable Detailed Monitoring** Showing data

Below are your CloudWatch metrics for the selected resources (a maximum of 10). Click on a graph to see an expanded view. A
▶ [View all CloudWatch metrics](#)

CPU Utilization (Percent) Disk Reads (Bytes) Disk Read Operations (Operations)

Enable Detailed Monitoring

×

Enable detailed monitoring for your instance to get these metrics at 1-minute frequency.
[Learn more](#)

Are you sure you want to enable detailed monitoring for the following instances?
(Additional charges apply.)

i-09990657415104108 (linux)

Cancel

Yes, Enable

Then Click on Yes Enable

Then go to Instance ---> Monitoring -----click on Create Alarm

Filter by tags and attributes or search by keyword

1 to 1 of 1

☐	Name	Instance ID	Instance Type	Availability Zone	Instance State	Status Checks	Alarm Status	Public DNS (If Available)
☐	linux	i-09990657415104108	t2.micro	ap-southeast-2a	running	2/2 checks ...	OK	ec2-13-210-109-155.ap-southeast-2.compute.amazonaws.com

Instance: i-09990657415104108 (linux)

Public DNS: ec2-13-210-109-155.ap-southeast-2.compute.amazonaws.com

Description

Status Checks

Monitoring

Tags

▶ CloudWatch alarms: 1 of 1 in OK

Create Alarm

CloudWatch metrics: Basic monitoring. [Enable Detailed Monitoring](#)

Showing data for: Last Hour

Below are your CloudWatch metrics for the selected resources (a maximum of 10). Click on a graph to see an expanded view. All times shown are in UTC.

View all CloudWatch metrics

Create Alarm

You can use CloudWatch alarms to be notified automatically whenever metric data reaches a level you define.

To edit an alarm, first choose whom to notify and then define when the notification should be sent.

☒ **Send a notification to:** Alarm (kavithab652@gmail.com) [create topic](#)

☒ **Take the action**

☐ Recover this instance 

☒ Stop this instance 

☐ Terminate this instance 

☐ Reboot this instance 

AWS will create the following Service Linked Role in your account to perform this EC2 action. [Learn more.](#)

AWSServiceRoleForCloudWatchEvents ([show IAM policy document](#))

Whenever: Average of CPU Utilization

Is:

For at least: 1 consecutive period(s) of 5 Minutes

Select the Take action Check box- Choose the option as per your requirement

Here we selected as Stop this instance

Set the threshold value

Alarm created successfully



Click the alarm to view additional details and options in Amazon CloudWatch (opens in a new window)

[awsec2-i-09990657415104108-CPU-Utilization](#)

Note: If you created a new SNS topic or added a new email address, each new address will receive a subscription email that must be confirmed within three days. Notifications will only be sent to confirmed addresses.

Close

Go to OS level (Putty)

Login with User id - ec2-user

Execute below commands

sudo -i

top

```
root@ip-10-0-0-4:~  
top - 11:29:03 up 3:27, 1 user, load average: 0.02, 0.01, 0.00  
Tasks: 83 total, 1 running, 47 sleeping, 0 stopped, 0 zombie  
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni, 100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st  
KiB Mem : 1009388 total, 617220 free, 59516 used, 332652 buff/cache  
KiB Swap: 0 total, 0 free, 0 used. 790364 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1	root	20	0	125548	5404	4068	S	0.0	0.5	0:01.40	systemd
2	root	20	0	0	0	0	S	0.0	0.0	0:00.00	kthreadd
4	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/0:0H
5	root	20	0	0	0	0	I	0.0	0.0	0:00.01	kworker/u30+
6	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	mm_percpu_wq
7	root	20	0	0	0	0	S	0.0	0.0	0:00.12	ksoftirqd/0
8	root	20	0	0	0	0	I	0.0	0.0	0:00.15	rcu_sched
9	root	20	0	0	0	0	I	0.0	0.0	0:00.00	rcu_bh
10	root	rt	0	0	0	0	S	0.0	0.0	0:00.00	migration/0
11	root	rt	0	0	0	0	S	0.0	0.0	0:00.03	watchdog/0
12	root	20	0	0	0	0	S	0.0	0.0	0:00.00	cpuhp/0
13	root	20	0	0	0	0	S	0.0	0.0	0:00.00	kdevtmpfs
14	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	netns
20	root	20	0	0	0	0	S	0.0	0.0	0:00.00	xenbus
21	root	20	0	0	0	0	S	0.0	0.0	0:00.01	xenwatch
170	root	20	0	0	0	0	S	0.0	0.0	0:00.00	khungtaskd
171	root	20	0	0	0	0	S	0.0	0.0	0:00.00	oom reaper

In the above screen Utilization of CPU is 0%

To increase the cpu utilization execute the below command to generate alarm

dd if=/dev/zero of=/dev/null

root@ip-10-0-0-4:~

```
top - 11:38:25 up 3:36, 2 users, load average: 0.15, 0.03, 0.01
Tasks: 89 total, 2 running, 52 sleeping, 0 stopped, 0 zombie
%Cpu(s): 25.0 us, 75.0 sy, 0.0 ni, 0.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
KiB Mem : 1009388 total, 610612 free, 65548 used, 333228 buff/cache
KiB Swap: 0 total, 0 free, 0 used. 783892 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
32648	root	20	0	114728	796	732	R	99.7	0.1	0:10.21	dd
1	root	20	0	123348	3404	4068	S	0.0	0.3	0:01.41	systemd
2	root	20	0	0	0	0	S	0.0	0.0	0:00.00	kthreadd
4	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/0:0H
5	root	20	0	0	0	0	I	0.0	0.0	0:00.01	kworker/u30:0
6	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	mm_percpu_wq
7	root	20	0	0	0	0	S	0.0	0.0	0:00.12	ksoftirqd/0
8	root	20	0	0	0	0	I	0.0	0.0	0:00.16	rcu_sched
9	root	20	0	0	0	0	I	0.0	0.0	0:00.00	rcu_bh

In the above screen CPU Utilization increased to 99.7%

We got the Alarm (alert) through mail

The screenshot shows a Gmail interface with an email titled "ALARM: 'cpu1' in Asia Pacific (Sydney)". The email content states that the Amazon CloudWatch Alarm "cpu1" has entered the ALARM state because the threshold was crossed. It includes details about the alarm, the reason for the state change, and a link to view the alarm in the AWS Management Console.

ALARM: "cpu1" in Asia Pacific (Sydney)

CPUSpike no-reply@sns.amazonaws.com via amazonses.com
to me

5:15 PM (1 minute ago)

You are receiving this email because your Amazon CloudWatch Alarm "cpu1" in the Asia Pacific (Sydney) region has entered the ALARM state, because "Threshold Crossed: 1 datapoint [99.6382513661202 (20/01/19 11:39:00)] was greater than or equal to the threshold (50.0)," at "Sunday 20 January, 2019 11:45:12 UTC".

View this alarm in the AWS Management Console:
<https://console.aws.amazon.com/cloudwatch/home?region=ap-southeast-2#s=Alarms&alarm=cpu1>

Alarm Details:

- Name: cpu1
- Description: cpu1
- State Change: INSUFFICIENT_DATA -> ALARM
- Reason for State Change: Threshold Crossed: 1 datapoint [99.6382513661202 (20/01/19 11:39:00)] was greater than or equal to the threshold (50.0).
- Timestamp: Sunday 20 January, 2019 11:45:12 UTC
- AWS Account: 996488761977

Threshold:

- The alarm is in the ALARM state when the metric is GreaterThanOrEqualToThreshold 50.0 for 60 seconds.

Monitored Metric:

Once the Alarm generated and the instance is stopped.

The screenshot shows the AWS Management Console interface for an EC2 instance. The instance is named "linux" and is in the "stopped" state. The "Instance State" column shows a red stop icon and the word "stopped". The "Alarm Status" column shows a red alarm icon and the word "ALARM".

Launch Instance Connect Actions

Filter by tags and attributes or search by keyword

Name	Instance ID	Instance Type	Availability Zone	Instance State	Status Checks	Alarm Status	Pub
linux	i-099906574151041...	t2.micro	ap-southeast-2a	stopped		ALARM	

